

ASHOK PASALA

Founder & CEO (Varellen Technologies) • Founder & Systems Architect (Norveth, OSPRED, QROS)

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EXECUTIVE SUMMARY

Founder, AI Systems Architect, and Deep Tech Engineer engineering autonomous multi-agent orchestration frameworks, quantitative ML trading systems, MCP discovery search engines, and high-concurrency distributed backends. Founder & CEO of **Varellen Technologies**, Architect of **Norveth** (norveth.app), **OSPRED**, and **QROS**. Builder of **40+ shipped software systems across 50+ global hackathons** (Google Cloud Rapid Agent, Bank of India & IIT Hyderabad, SanDisk, Qwen Global). Computer Science & Distributed Systems at **VIT-AP University** (2024–2028).

VENTURE LEADERSHIP & SYSTEMS ENGINEERING

Varellen Technologies — Founder & Chief Executive Officer 2024 – PRESENT | ENTERPRISE & SYSTEMS

- Guiding venture roadmap and system design for enterprise digital platforms, closed-loop AI automation pipelines, and high-throughput cloud infrastructure.
- Architected fault-tolerant microservice meshes with sub-millisecond API dispatch, zero-trust security postures, and automated Docker/K8s CI/CD pipelines.

Norveth — Founder & Systems Architect (norveth.app) 2024 – PRESENT | ENGINEERING INTELLIGENCE

- Architected full-lifecycle website engineering intelligence platform utilizing **Playwright dynamic headless crawling** with exponential backoff & LLM failover, **Model Context Protocol (MCP)** tooling, and automated OWASP Top 10 security/performance diagnostics.
- Engineered Engineering Knowledge Graph reasoning layer mapping findings into interactive architecture diagrams and deterministic code fixes.

OSPRED & Yaltan — Creator & Systems Architect 2024 – PRESENT | AI CAPABILITY SEARCH

- Engineered autonomous discovery and indexing engine for Model Context Protocol (MCP) servers, reusable AI skills, and plugins using the Yaltan scraping pipeline.
- Designed Canonical Capability Schema and automated tool verification engine with SSRF/prompt-injection safety metadata scoring.

QROS Quantitative Research — Quantitative Systems Architect 2024 – PRESENT | QUANT ML TRADING

- Built high-frequency trading platform targeting **Polymarket BTC 5-minute prediction markets** using Binance 1-second streaming tick feeds.
- Trained LightGBM Scale/Shape ML models predicting strike settlement probabilities ($p_{up} = P(S_T > K)$) and volatility alongside an MM-1 market-making engine.

FLAGSHIP ARCHITECTURES & HACKATHON SYSTEMS (50+ HACKATHONS)

SlackMind Agent Command Center Google Cloud Rapid Agent Gemini, Vertex AI Data Store, MCP, Slack Bolt, Python

- Autonomous AI agent orchestration kernel coordinating intelligent agent swarms directly inside Slack with Gemini models, Vertex AI Search Data Store, and MCP tool routing.

The Vector / Vector AI Command Center Autonomous C-Suite Multi-Agent Swarms, Python, LangChain, FastAPI, Next.js

- Autonomous multi-agent executive council (CEO, CTO, CMO) that transforms startup concepts into complete business plans, distributed tech architectures, and financial models with cross-agent consensus in seconds.

Mule Account Detection System Bank of India & IIT Hyderabad Graph ML, Anomaly Detection, Python, FastAPI, FinTech

- High-frequency financial crime and suspicious transaction detection system identifying illicit mule account networks in real-time.

IWCIS Storage Architecture & Qwen Chaos SanDisk & Qwen Global exFAT Internals, Container Architecture, Qwen LLM, Vector Memory

- Containerized exFAT file-system storage optimization (SanDisk) paired with a self-persistent neural MemoryAgent with cross-session state recall (Qwen Global).

TECHNICAL ARSENAL & DOMAIN MASTERY

AUTONOMOUS AI & QUANT: Multi-Agent Swarms, Model Context Protocol (MCP), LightGBM, Quantitative Volatility, LangChain, RAG Pipelines, Vector DBs (ChromaDB, Pinecone).

LANGUAGES: Python, TypeScript, JavaScript, Go, Rust, Java, C/C++, SQL, Bash/Shell, Solidity.

BACKEND & INFRA: FastAPI, Node.js, Express, Next.js 16 (App Router), React 19, PostgreSQL, Redis Streams, Docker, Kubernetes, Playwright, Cloudflare Tunnels.

SECURITY & WEB3: OWASP Top 10 Hardening, SSRF Protection, Zero-Trust Architecture, EVM Smart Contracts, Cryptographic Protocols, Graph Anomaly Detection.

EDUCATION & 50+ HACKATHONS TRACK RECORD

VIT-AP University — B.Tech in Computer Science & Engineering 2024 – 2028 | ANDHRA PRADESH, INDIA

Core Focus: Distributed Systems, Operating Systems Internals, Advanced Algorithms, Computer Networks, Database Architecture, Machine Learning.

50+ Global Hackathons & Challenges: Google Cloud Rapid Agent, Bank of India & IIT Hyderabad, SanDisk Storage, Qwen Global, Slack AI, Google Devpost/XPRIZE.